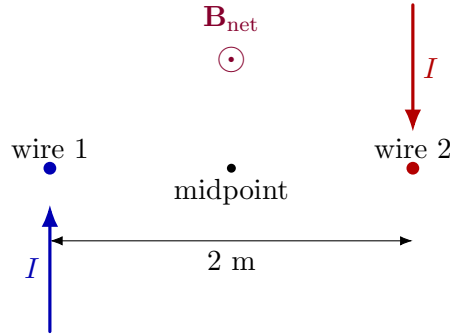


1. A 50 kV direct-current power line consists of two wire conductors 2 m apart. When this line is transmitting 10 MW of power, how strong is the magnetic field midway between the conductors?

Solution



Step 1: Find the current in the line

The transmitted power is

$$P = VI.$$

Given:

$$P = 10 \text{ MW} = 10 \times 10^6 \text{ W},$$

$$V = 50 \text{ kV} = 50 \times 10^3 \text{ V}.$$

Therefore,

$$I = \frac{P}{V} = \frac{10 \times 10^6}{50 \times 10^3} = 200 \text{ A}.$$

So each conductor carries a current of magnitude

$$I = 200 \text{ A}.$$

Step 2: Distance of the midpoint from each wire

The two conductors are 2 m apart, so the midpoint is

$$r = 1 \text{ m}$$

from each wire.

Step 3: Magnetic field due to one long straight wire

For a long straight wire,

$$B = \frac{\mu_0 I}{2\pi r}.$$

Substituting

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}, \quad I = 200 \text{ A}, \quad r = 1 \text{ m},$$

we get

$$B_1 = \frac{4\pi \times 10^{-7} \times 200}{2\pi \times 1} \quad (1)$$

$$= 4 \times 10^{-5} \text{ T}. \quad (2)$$

Hence the magnetic field due to one conductor at the midpoint is

$$\boxed{B_1 = 4 \times 10^{-5} \text{ T}}.$$

Step 4: Add the contributions from the two wires

Since the currents in the two conductors are equal and opposite, the magnetic fields at the midpoint are in the same direction (by the right-hand rule), so they add.

Therefore,

$$B_{\text{net}} = B_1 + B_1 = 2B_1.$$

So

$$B_{\text{net}} = 2(4 \times 10^{-5}) = 8 \times 10^{-5} \text{ T}.$$

Thus,

$$\boxed{B_{\text{mid}} = 8 \times 10^{-5} \text{ T}}.$$

Step 5: Express in microtesla

Since

$$1 \mu\text{T} = 10^{-6} \text{ T},$$

we get

$$8 \times 10^{-5} \text{ T} = 80 \mu\text{T}.$$

So the answer may also be written as

$$\boxed{B_{\text{mid}} = 80 \mu\text{T}}.$$

Final Answer

The magnetic field midway between the conductors is

$$8 \times 10^{-5} \text{ T}$$

or equivalently,

$$80 \mu\text{T}.$$

2. Consider two circular rings of copper wire. One ring is a scaled-up version of the other, twice as large in all regards (radius, crosssectional radius). If currents around the rings are driven by equal voltage sources, how do the magnetic fields at the centers compare?

Solution

Given idea

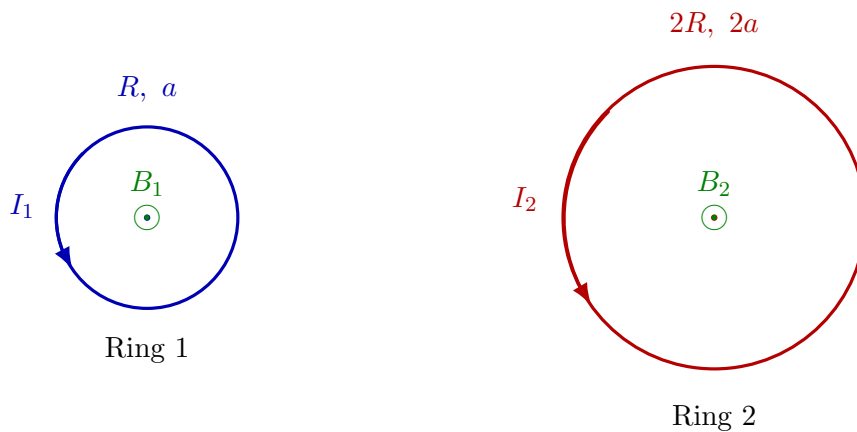
We compare two circular copper rings:

- **Ring 1:** smaller ring, radius R , wire cross-sectional radius a .
- **Ring 2:** a geometrically scaled version, **twice as large in every linear dimension**, so:

$$R' = 2R, \quad a' = 2a.$$

Both are made of the same material (copper), and each is connected to an **equal voltage source** V .

We want to compare the magnetic fields at the centers of the two rings.



Step 1: Magnetic field at the center of a circular loop

For a circular loop of radius R carrying current I , the magnetic field at the center is

$$B = \frac{\mu_0 I}{2R}.$$

Therefore, to compare the magnetic fields, we need to compare the ratio I/R for the two rings.

Step 2: Find how the electrical resistance scales

The resistance of a wire is

$$\mathcal{R} = \rho \frac{L}{A},$$

where

- ρ is the resistivity of copper,
- L is the length of the wire,
- A is the cross-sectional area of the wire.

For Ring 1: Its wire length is the circumference:

$$L_1 = 2\pi R.$$

Its cross-sectional area is

$$A_1 = \pi a^2.$$

So its resistance is

$$\mathcal{R}_1 = \rho \frac{2\pi R}{\pi a^2} = \rho \frac{2R}{a^2}.$$

For Ring 2: Since every linear dimension is doubled,

$$R_2 = 2R, \quad a_2 = 2a.$$

Hence its length is

$$L_2 = 2\pi(2R) = 4\pi R = 2L_1,$$

and its cross-sectional area is

$$A_2 = \pi(2a)^2 = 4\pi a^2 = 4A_1.$$

Therefore,

$$\mathcal{R}_2 = \rho \frac{L_2}{A_2} \quad (3)$$

$$= \rho \frac{2L_1}{4A_1} \quad (4)$$

$$= \frac{1}{2} \rho \frac{L_1}{A_1} \quad (5)$$

$$= \frac{\mathcal{R}_1}{2}. \quad (6)$$

So the larger ring has only half the resistance of the smaller ring:

$$\boxed{\mathcal{R}_2 = \frac{\mathcal{R}_1}{2}}$$

Step 3: Compare the currents for equal applied voltage

Each ring is connected to the same voltage V . By Ohm's law,

$$I = \frac{V}{\mathcal{R}}.$$

Thus,

$$I_1 = \frac{V}{\mathcal{R}_1}, \quad I_2 = \frac{V}{\mathcal{R}_2}.$$

Since $\mathcal{R}_2 = \mathcal{R}_1/2$,

$$I_2 = \frac{V}{\mathcal{R}_1/2} = 2 \frac{V}{\mathcal{R}_1} = 2I_1.$$

Hence the current in the larger ring is twice the current in the smaller ring:

$$\boxed{I_2 = 2I_1}$$

Step 4: Compare the magnetic fields

Now apply the center-field formula.

For Ring 1:

$$B_1 = \frac{\mu_0 I_1}{2R}.$$

For Ring 2, whose radius is $2R$ and current is $2I_1$,

$$B_2 = \frac{\mu_0 I_2}{2(2R)} = \frac{\mu_0 (2I_1)}{4R} = \frac{\mu_0 I_1}{2R}.$$

Therefore,

$$B_2 = B_1$$

So the magnetic fields at the centers are **equal**.

Physical interpretation

Doubling the size of the ring has two competing effects:

- the radius doubles, which by itself would reduce the field at the center by a factor of 2,
- but the wire also becomes thicker, so the resistance drops by a factor of 2, and with the same applied voltage the current doubles, which increases the field by a factor of 2.

These two effects exactly cancel.

Final answer

Final Result

If one circular copper ring is scaled up by a factor of 2 in *all* linear dimensions (both loop radius and wire radius), and both rings are driven by equal voltages, then

$$B_{\text{large}} = B_{\text{small}}$$

That is, **the magnetic fields at the centers are the same.**

-
3. (a) Find H in rectangular components at $P(2, 3, 4)$ if there is a current filament on the z axis carrying 8 mA in the $\hat{a}_z$ direction. (b) Repeat if the filament is located at $x = -1, y = 2$. (c) Find H if both filaments are present. (Hint: $\vec{B} = \mu_0 \vec{H}$)

Solution

Since each filament is directed along $+\hat{\mathbf{a}}_z$, the magnetic field circles the filament according to the right-hand rule. Also, because the filaments are infinite and parallel to the z -axis, the field is independent of z . So for the point $P(2, 3, 4)$, only the x - and y -coordinates matter.

General formula for a z -directed current filament

For a filament carrying current I along $+\hat{\mathbf{a}}_z$, located at (x_0, y_0) , define the transverse displacement from the filament to the field point (x, y) as

$$\boldsymbol{\rho} = (x - x_0)\hat{\mathbf{a}}_x + (y - y_0)\hat{\mathbf{a}}_y.$$

Then

$$\rho^2 = (x - x_0)^2 + (y - y_0)^2.$$

Using

$$\mathbf{H} = \frac{I}{2\pi\rho} \hat{\mathbf{a}}_\phi$$

and

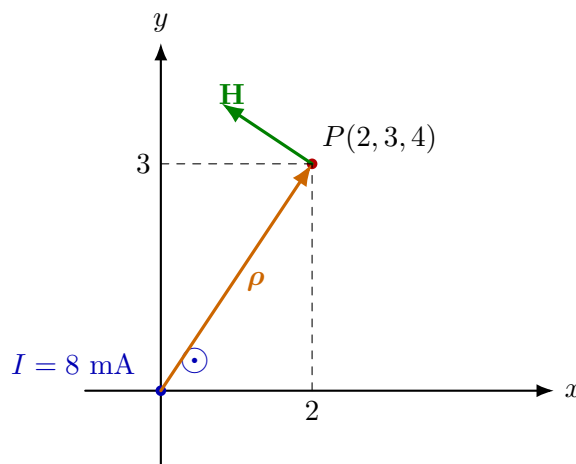
$$\hat{\mathbf{a}}_\phi = \frac{\hat{\mathbf{a}}_z \times \boldsymbol{\rho}}{\rho},$$

we get the rectangular-component form

$$\mathbf{H} = \frac{I}{2\pi[(x - x_0)^2 + (y - y_0)^2]} [-(y - y_0)\hat{\mathbf{a}}_x + (x - x_0)\hat{\mathbf{a}}_y].$$

Here

$$I = 8 \text{ mA} = 8 \times 10^{-3} \text{ A}.$$



(a) Filament on the z -axis

Here

$$x_0 = 0, \quad y_0 = 0.$$

At $P(2, 3, 4)$,

$$x - x_0 = 2, \quad y - y_0 = 3,$$

so

$$\rho_1^2 = 2^2 + 3^2 = 13.$$

Therefore

$$\mathbf{H}_1 = \frac{8 \times 10^{-3}}{2\pi(13)} (-3\hat{\mathbf{a}}_x + 2\hat{\mathbf{a}}_y) \quad (7)$$

$$= \frac{4 \times 10^{-3}}{13\pi} (-3\hat{\mathbf{a}}_x + 2\hat{\mathbf{a}}_y) \text{ A/m.} \quad (8)$$

Hence

$$\mathbf{H}_1 = (-2.94 \times 10^{-4}\hat{\mathbf{a}}_x + 1.96 \times 10^{-4}\hat{\mathbf{a}}_y) \text{ A/m.}$$

There is no z -component:

$$H_{1z} = 0.$$

(b) Filament located at $x = -1, y = 2$

Now

$$x_0 = -1, \quad y_0 = 2.$$

So at $P(2, 3, 4)$,

$$x - x_0 = 2 - (-1) = 3, \quad y - y_0 = 3 - 2 = 1.$$

Thus

$$\rho_2^2 = 3^2 + 1^2 = 10.$$

Therefore

$$\mathbf{H}_2 = \frac{8 \times 10^{-3}}{2\pi(10)} (-1\hat{\mathbf{a}}_x + 3\hat{\mathbf{a}}_y) \quad (9)$$

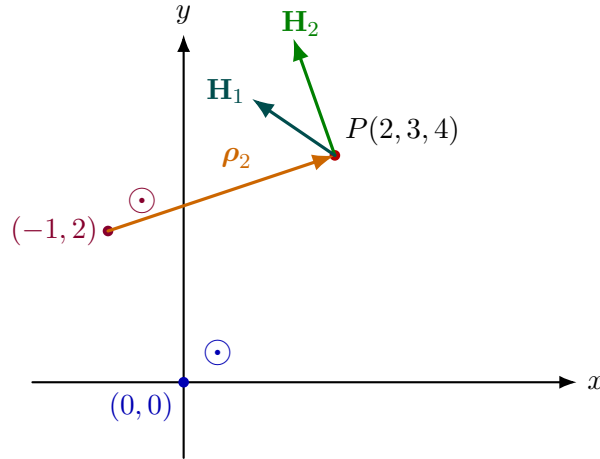
$$= \frac{4 \times 10^{-4}}{\pi} (-\hat{\mathbf{a}}_x + 3\hat{\mathbf{a}}_y) \text{ A/m.} \quad (10)$$

Hence

$$\mathbf{H}_2 = (-1.27 \times 10^{-4}\hat{\mathbf{a}}_x + 3.82 \times 10^{-4}\hat{\mathbf{a}}_y) \text{ A/m.}$$

Again,

$$H_{2z} = 0.$$



(c) Both filaments present

By superposition,

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2.$$

So

$$\mathbf{H} = (-2.94 \times 10^{-4} \hat{\mathbf{a}}_x + 1.96 \times 10^{-4} \hat{\mathbf{a}}_y) \quad (11)$$

$$+ (-1.27 \times 10^{-4} \hat{\mathbf{a}}_x + 3.82 \times 10^{-4} \hat{\mathbf{a}}_y). \quad (12)$$

Therefore,

$$\mathbf{H} = (-4.21 \times 10^{-4} \hat{\mathbf{a}}_x + 5.78 \times 10^{-4} \hat{\mathbf{a}}_y) \text{ A/m.}$$

and

$$H_z = 0.$$

So the final answers are:

Final Answers

$$\mathbf{H}_1 = (-2.94 \times 10^{-4} \hat{\mathbf{a}}_x + 1.96 \times 10^{-4} \hat{\mathbf{a}}_y) \text{ A/m}$$

$$\mathbf{H}_2 = (-1.27 \times 10^{-4} \hat{\mathbf{a}}_x + 3.82 \times 10^{-4} \hat{\mathbf{a}}_y) \text{ A/m}$$

$$\mathbf{H} = (-4.21 \times 10^{-4} \hat{\mathbf{a}}_x + 5.78 \times 10^{-4} \hat{\mathbf{a}}_y) \text{ A/m}$$

4. The parallel filamentary conductors shown in Figure 1 lie in free space. Plot $|H|$ versus y , $4 < y < 4$, along the line $x = 0$, $z = 2$.

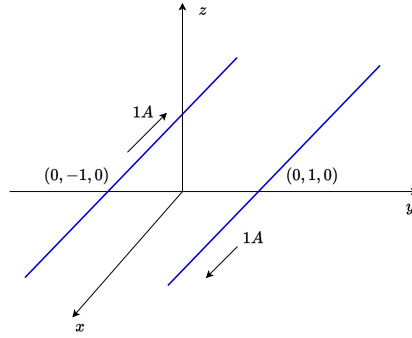
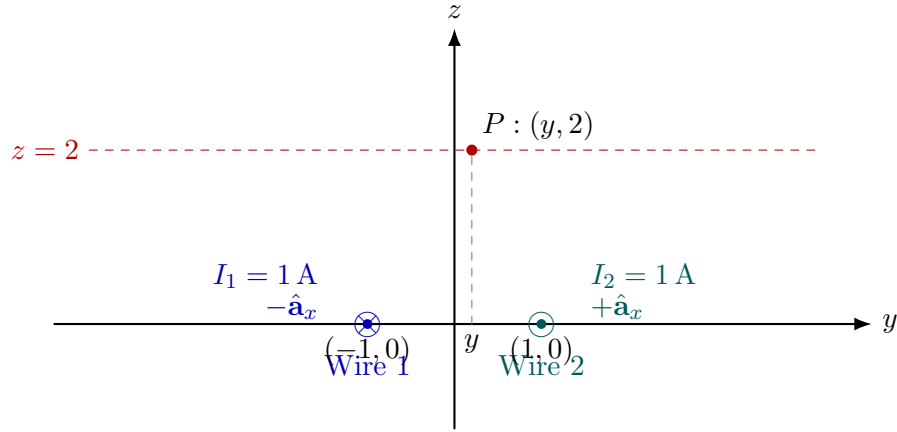


Figure 1: Q4

Solution



1. Geometry and current directions

From the figure, the two infinite filamentary conductors are parallel to the x -axis and pass through

$$(0, -1, 0) \quad \text{and} \quad (0, 1, 0).$$

Their currents are

$$I_1 = 1 \text{ A in } -\hat{\mathbf{a}}_x, \quad I_2 = 1 \text{ A in } +\hat{\mathbf{a}}_x.$$

We are asked to find $|\mathbf{H}|$ along the line

$$x = 0, \quad z = 2, \quad -4 < y < 4.$$

So the observation point is

$$P = (0, y, 2).$$

2. Field of an infinite filament parallel to the x -axis

For an infinite current filament carrying current I along $+\hat{\mathbf{a}}_x$, located at (y_0, z_0) , the magnetic field is

$$\mathbf{H} = \frac{I}{2\pi\rho^2} [-(z - z_0)\hat{\mathbf{a}}_y + (y - y_0)\hat{\mathbf{a}}_z],$$

where

$$\rho^2 = (y - y_0)^2 + (z - z_0)^2.$$

If the current is along $-\hat{\mathbf{a}}_x$, the sign reverses.

3. Field due to filament 1 at $(0, -1, 0)$

At the point $P = (0, y, 2)$,

$$y_0 = -1, \quad z_0 = 0,$$

so

$$y - y_0 = y + 1, \quad z - z_0 = 2,$$

and

$$\rho_1^2 = (y + 1)^2 + 4.$$

Since this filament carries 1 A in the $-\hat{\mathbf{a}}_x$ direction,

$$\mathbf{H}_1 = -\frac{1}{2\pi[(y + 1)^2 + 4]} [-2\hat{\mathbf{a}}_y + (y + 1)\hat{\mathbf{a}}_z].$$

Therefore,

$$\boxed{\mathbf{H}_1 = \frac{1}{2\pi[(y + 1)^2 + 4]} [2\hat{\mathbf{a}}_y - (y + 1)\hat{\mathbf{a}}_z].}$$

4. Field due to filament 2 at (0, 1, 0)

Now

$$y_0 = 1, \quad z_0 = 0,$$

so

$$y - y_0 = y - 1, \quad z - z_0 = 2,$$

and

$$\rho_2^2 = (y - 1)^2 + 4.$$

Since this filament carries 1 A in the $+\hat{\mathbf{a}}_x$ direction,

$$\mathbf{H}_2 = \frac{1}{2\pi[(y - 1)^2 + 4]} [-2\hat{\mathbf{a}}_y + (y - 1)\hat{\mathbf{a}}_z].$$

So,

$$\mathbf{H}_2 = \frac{1}{2\pi[(y - 1)^2 + 4]} [-2\hat{\mathbf{a}}_y + (y - 1)\hat{\mathbf{a}}_z].$$

5. Total magnetic field

By superposition,

$$\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2.$$

Hence,

$$\mathbf{H} = \frac{1}{2\pi} \left[\frac{2}{(y + 1)^2 + 4} - \frac{2}{(y - 1)^2 + 4} \right] \hat{\mathbf{a}}_y \quad (13)$$

$$+ \frac{1}{2\pi} \left[-\frac{y + 1}{(y + 1)^2 + 4} + \frac{y - 1}{(y - 1)^2 + 4} \right] \hat{\mathbf{a}}_z. \quad (14)$$

Now simplify the two components.

The y -component:

$$H_y = \frac{1}{2\pi} \left[\frac{2}{(y + 1)^2 + 4} - \frac{2}{(y - 1)^2 + 4} \right] = -\frac{4y}{\pi(y^4 + 6y^2 + 25)}.$$

The z -component:

$$H_z = \frac{1}{2\pi} \left[-\frac{y+1}{(y+1)^2+4} + \frac{y-1}{(y-1)^2+4} \right] = \frac{y^2-5}{\pi(y^4+6y^2+25)}.$$

Therefore,

$$\mathbf{H}(y) = -\frac{4y}{\pi(y^4+6y^2+25)} \hat{\mathbf{a}}_y + \frac{y^2-5}{\pi(y^4+6y^2+25)} \hat{\mathbf{a}}_z \quad \text{A/m.}$$

6. Magnitude of $\mathbf{H}$

Now

$$|\mathbf{H}| = \sqrt{H_y^2 + H_z^2}.$$

So

$$|\mathbf{H}| = \frac{1}{\pi(y^4+6y^2+25)} \sqrt{16y^2 + (y^2-5)^2} \quad (15)$$

$$= \frac{1}{\pi(y^4+6y^2+25)} \sqrt{y^4+6y^2+25}. \quad (16)$$

Hence,

$$|\mathbf{H}(y)| = \frac{1}{\pi\sqrt{y^4+6y^2+25}} \quad \text{A/m,} \quad -4 < y < 4.$$

This is the function to be plotted.

7. Some useful values

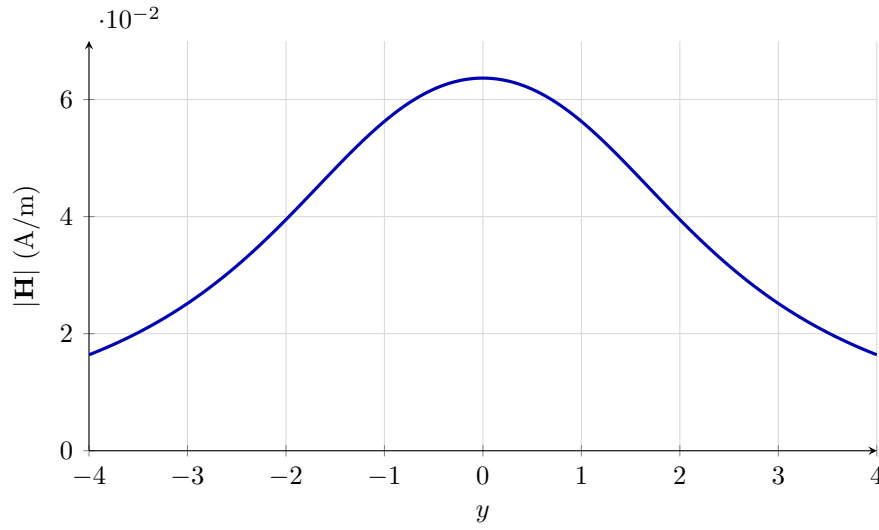
$$|\mathbf{H}(0)| = \frac{1}{5\pi} \approx 6.37 \times 10^{-2} \text{ A/m}$$

$$|\mathbf{H}(\pm 1)| = \frac{1}{\pi\sqrt{32}} \approx 5.63 \times 10^{-2} \text{ A/m}$$

$$|\mathbf{H}(\pm 4)| = \frac{1}{\pi\sqrt{377}} \approx 1.64 \times 10^{-2} \text{ A/m}$$

So the field magnitude is maximum near $y = 0$ and decreases symmetrically as $|y|$ increases.

8. Plot of $|\mathbf{H}|$ versus y



Final Answer

Along the line $x = 0$, $z = 2$, the magnetic field is

$$\mathbf{H}(y) = -\frac{4y}{\pi(y^4 + 6y^2 + 25)} \hat{\mathbf{a}}_y + \frac{y^2 - 5}{\pi(y^4 + 6y^2 + 25)} \hat{\mathbf{a}}_z \quad \text{A/m}$$

and its magnitude is

$$|\mathbf{H}(y)| = \frac{1}{\pi\sqrt{y^4 + 6y^2 + 25}} \quad \text{A/m}, \quad -4 < y < 4.$$

5. An infinite filament on the z axis carries 20π mA in the $\hat{\mathbf{a}}_z$ direction. Three $\hat{\mathbf{a}}_z$ -directed uniform cylindrical current sheets are also present: 400 mA/m at $\rho = 1$ cm, 250 mA/m at $\rho = 2$ cm, and 300 mA/m at $\rho = 3$ cm. Calculate H_ϕ at $\rho = 0.5$, 1.5, 2.5, and 3.5 cm.

Solution

1. Symmetry and Ampère's law

Because all currents are along the z -direction and the geometry is cylindrically symmetric about the z -axis, the magnetic field must be purely circumferential:

$$\mathbf{H} = H_\phi(\rho) \hat{\mathbf{a}}_\phi.$$

Using Ampère's law on a circular Amperian path of radius ρ ,

$$\oint \mathbf{H} \cdot d\mathbf{l} = I_{\text{enc}}.$$

Since $\mathbf{H}$ is tangential and constant in magnitude on the circle,

$$H_\phi(\rho) (2\pi\rho) = I_{\text{enc}}.$$

Hence,

$$H_\phi(\rho) = \frac{I_{\text{enc}}}{2\pi\rho}$$

So the whole problem reduces to finding the *net enclosed current* at each radius.

2. Convert each source into enclosed current

(i) Filament current

The filament at the origin carries

$$I_f = 20\pi \text{ mA} = 20\pi \times 10^{-3} \text{ A} = 0.02\pi \text{ A}.$$

(ii) Cylindrical current sheets

A cylindrical sheet at $\rho = a$ with surface current density $\mathbf{K} = K_z \hat{\mathbf{a}}_z$ contributes total current

$$I_{\text{sheet}} = K_z \times (\text{circumference}) = K_z(2\pi a).$$

Sheet at $\rho = 1 \text{ cm} = 0.01 \text{ m}$:

$$K_1 = 400 \text{ mA/m} = 0.4 \text{ A/m}.$$

So

$$I_1 = K_1(2\pi a_1) = 0.4(2\pi)(0.01) = 0.008\pi \text{ A}.$$

Sheet at $\rho = 2 \text{ cm} = 0.02 \text{ m}$:

$$K_2 = -250 \text{ mA/m} = -0.25 \text{ A/m}.$$

So

$$I_2 = K_2(2\pi a_2) = (-0.25)(2\pi)(0.02) = -0.01\pi \text{ A}.$$

Sheet at $\rho = 3 \text{ cm} = 0.03 \text{ m}$:

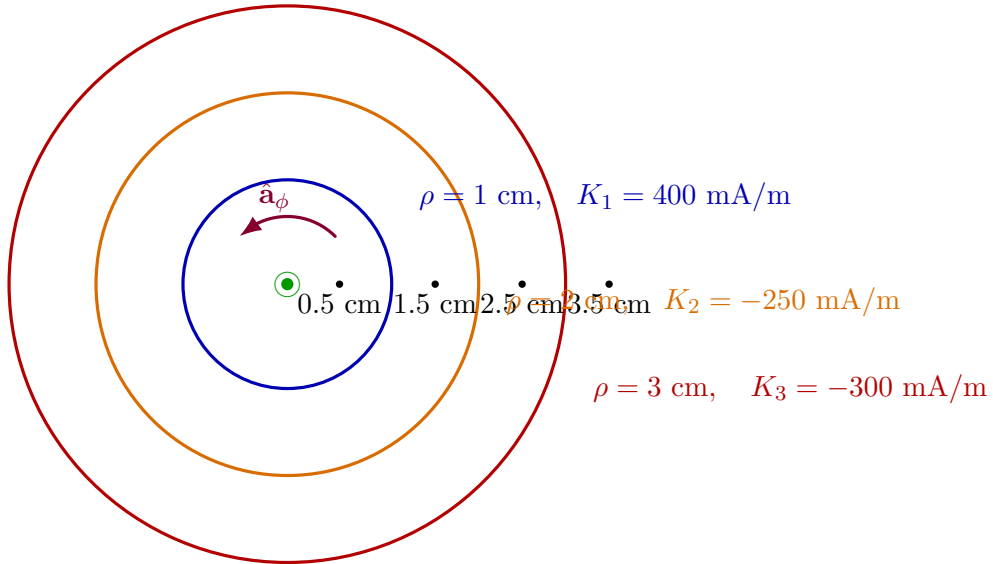
$$K_3 = -300 \text{ mA/m} = -0.3 \text{ A/m}.$$

So

$$I_3 = K_3(2\pi a_3) = (-0.3)(2\pi)(0.03) = -0.018\pi \text{ A}.$$

Thus,

$$\boxed{I_f = 0.02\pi \text{ A}}, \quad \boxed{I_1 = 0.008\pi \text{ A}}, \quad \boxed{I_2 = -0.01\pi \text{ A}}, \quad \boxed{I_3 = -0.018\pi \text{ A}}.$$



3. Find H_ϕ in each region

Region 1: $\rho < 1 \text{ cm}$

Only the central filament is enclosed, so

$$I_{\text{enc}} = I_f = 0.02\pi \text{ A}.$$

Hence,

$$H_\phi(\rho) = \frac{0.02\pi}{2\pi\rho} = \frac{0.01}{\rho} \quad (\rho \text{ in meters}).$$

Region 2: $1 < \rho < 2$ cm

Now the filament and the first sheet are enclosed:

$$I_{\text{enc}} = I_f + I_1 = (0.02\pi + 0.008\pi) = 0.028\pi \text{ A.}$$

Therefore,

$$H_\phi(\rho) = \frac{0.028\pi}{2\pi\rho} = \frac{0.014}{\rho}.$$

Region 3: $2 < \rho < 3$ cm

Now the filament and the first two sheets are enclosed:

$$I_{\text{enc}} = I_f + I_1 + I_2 = 0.02\pi + 0.008\pi - 0.01\pi = 0.018\pi \text{ A.}$$

So,

$$H_\phi(\rho) = \frac{0.018\pi}{2\pi\rho} = \frac{0.009}{\rho}.$$

Region 4: $\rho > 3$ cm

Now all currents are enclosed:

$$I_{\text{enc}} = I_f + I_1 + I_2 + I_3 = 0.02\pi + 0.008\pi - 0.01\pi - 0.018\pi = 0.$$

Therefore,

$$H_\phi(\rho) = 0.$$

4. Evaluate at the requested radii

(a) At $\rho = 0.5 \text{ cm} = 0.005 \text{ m}$

This lies in Region 1, so

$$H_\phi = \frac{0.01}{0.005} = 2 \text{ A/m.}$$

$$H_\phi(0.5 \text{ cm}) = 2 \text{ A/m}$$

(b) At $\rho = 1.5 \text{ cm} = 0.015 \text{ m}$

This lies in Region 2, so

$$H_\phi = \frac{0.014}{0.015} = 0.9333 \text{ A/m.}$$

$$H_\phi(1.5 \text{ cm}) = 0.933 \text{ A/m}$$

(c) At $\rho = 2.5 \text{ cm} = 0.025 \text{ m}$

This lies in Region 3, so

$$H_\phi = \frac{0.009}{0.025} = 0.36 \text{ A/m.}$$

$$H_\phi(2.5 \text{ cm}) = 0.36 \text{ A/m}$$

(d) At $\rho = 3.5 \text{ cm} = 0.035 \text{ m}$

This lies in Region 4, so

$$H_\phi = 0.$$

$$H_\phi(3.5 \text{ cm}) = 0$$

5. Piecewise expression

It is also useful to summarize the field as

$$H_{\phi}(\rho) = \begin{cases} \frac{0.01}{\rho}, & 0 < \rho < 0.01 \text{ m}, \\ \frac{0.014}{\rho}, & 0.01 < \rho < 0.02 \text{ m}, \\ \frac{0.009}{\rho}, & 0.02 < \rho < 0.03 \text{ m}, \\ 0, & \rho > 0.03 \text{ m}. \end{cases}$$

where ρ is in meters and H_{ϕ} is in A/m.

Final Answers

$$H_{\phi}(0.5 \text{ cm}) = 2 \text{ A/m}$$

$$H_{\phi}(1.5 \text{ cm}) = 0.933 \text{ A/m}$$

$$H_{\phi}(2.5 \text{ cm}) = 0.36 \text{ A/m}$$

$$H_{\phi}(3.5 \text{ cm}) = 0$$

6. A square wire frame with side length ℓ has total resistance R . It is being pulled with speed v out of a region where there is a uniform magnetic field $\mathbf{B}$ pointing out of the page (the shaded area in Fig. 2). Consider the moment when the left corner of the square is a distance x inside the shaded area.
- What force do you need to apply to the square so that it moves with constant speed v ?
 - Verify that the work you do from $x = x_0$ (which you can assume is less than $\ell/\sqrt{2}$) down to $x = 0$ equals the energy dissipated in the resistor.

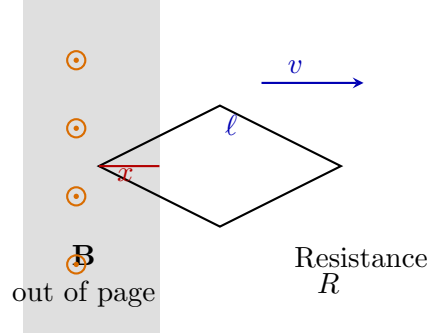


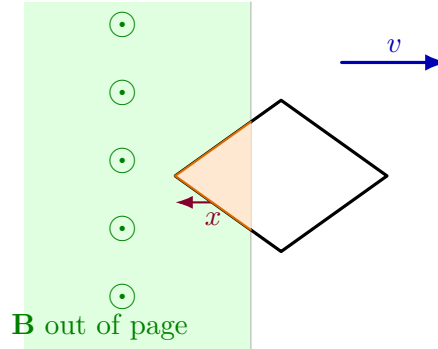
Figure 2: A square conducting frame being pulled out of a region of uniform magnetic field.

Solution

For the interval specified in the problem,

$$0 < x < \frac{\ell}{\sqrt{2}},$$

the magnetic-field boundary cuts only the *left corner* portion of the square. Hence the part of the loop inside the field is a small isosceles triangle.



1. Area of the loop inside the field

Because the square is tilted by 45° , its left two sides make $\pm 45^\circ$ with the horizontal. So if the left corner is a distance x inside the field, the vertical cut at the field boundary intercepts those two sides at heights $\pm x$. Therefore the portion inside the field is a triangle of

- horizontal height x ,
- vertical base $2x$.

Hence the area inside the field is

$$A(x) = \frac{1}{2}(2x)(x) = x^2.$$

So the magnetic flux through the loop is

$$\Phi_B(x) = B A(x) = Bx^2.$$

2. Induced emf and current

As the frame is pulled to the right with constant speed v , the distance x decreases:

$$\frac{dx}{dt} = -v.$$

Therefore,

$$\mathcal{E} = \left| -\frac{d\Phi_B}{dt} \right| = \left| -\frac{d}{dt}(Bx^2) \right| \quad (17)$$

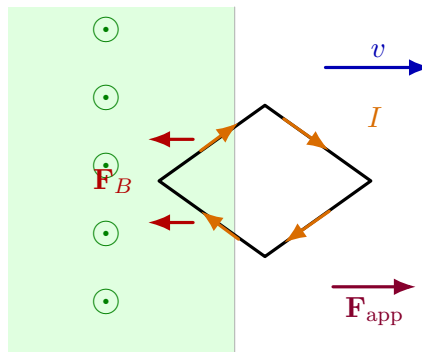
$$= \left| -2Bx \frac{dx}{dt} \right| \quad (18)$$

$$= 2Bxv. \quad (19)$$

Thus the induced current magnitude is

$$I(x) = \frac{\mathcal{E}}{R} = \frac{2Bxv}{R}.$$

Direction of current: As the loop leaves the field region, the outward magnetic flux is decreasing. By Lenz's law, the induced current must try to maintain outward flux, so the induced current is *counterclockwise*.



3. Magnetic force on the loop

Only the two left segments of the square lie inside the field. Each such segment has length

$$L_{\text{in}} = \sqrt{x^2 + x^2} = x\sqrt{2}.$$

For one segment, the magnetic force magnitude is

$$F_1 = I L_{\text{in}} B = I(x\sqrt{2})B.$$

This force is perpendicular to the segment. Since the segment is at 45° , the horizontal component is

$$F_{1x} = F_1 \cos 45^\circ = I(x\sqrt{2})B \left(\frac{1}{\sqrt{2}} \right) = IBx.$$

The lower left segment gives the same horizontal component, also opposing the motion. Therefore the total magnetic drag force is

$$F_B = 2IBx.$$

Substitute $I = \frac{2Bxv}{R}$:

$$F_B = 2 \left(\frac{2Bxv}{R} \right) Bx = \frac{4B^2 x^2 v}{R}.$$

So the magnetic force on the frame is to the left, and to maintain constant speed we must apply an equal rightward force:

$$F_{\text{app}} = \frac{4B^2 x^2 v}{R} \quad (\text{to the right}).$$

4. Work done from $x = x_0$ to $x = 0$

As the frame moves rightward, x decreases. A rightward displacement ds corresponds to

$$ds = -dx.$$

Hence the work done by the applied force is

$$W_{\text{app}} = \int F_{\text{app}} ds \quad (20)$$

$$= \int_{x_0}^0 F_{\text{app}}(-dx) \quad (21)$$

$$= \int_0^{x_0} \frac{4B^2 x^2 v}{R} dx \quad (22)$$

$$= \frac{4B^2 v}{R} \int_0^{x_0} x^2 dx \quad (23)$$

$$= \frac{4B^2 v}{R} \left(\frac{x_0^3}{3} \right). \quad (24)$$

Therefore,

$$W_{\text{app}} = \frac{4B^2 v x_0^3}{3R}.$$

5. Energy dissipated in the resistor

The Joule heating is

$$W_R = \int I^2 R dt.$$

Using

$$I = \frac{2Bxv}{R},$$

we get

$$I^2 R = \left(\frac{2Bxv}{R} \right)^2 R = \frac{4B^2 x^2 v^2}{R}.$$

So

$$W_R = \int \frac{4B^2 x^2 v^2}{R} dt. \quad (25)$$

Now use

$$dt = \frac{-dx}{v}.$$

Hence

$$W_R = \int_{x_0}^0 \frac{4B^2 x^2 v^2}{R} \left(\frac{-dx}{v} \right) \quad (26)$$

$$= \frac{4B^2 v}{R} \int_0^{x_0} x^2 dx \quad (27)$$

$$= \frac{4B^2 v}{R} \left(\frac{x_0^3}{3} \right). \quad (28)$$

Therefore,

$$W_R = \frac{4B^2 v x_0^3}{3R}.$$

So indeed,

$$W_{\text{app}} = W_R.$$

Final Answers

For $0 < x < \ell/\sqrt{2}$,

$$\mathcal{E} = 2Bxv, \quad I = \frac{2Bxv}{R}$$

$$F_{\text{app}} = \frac{4B^2 x^2 v}{R} \quad (\text{to the right, to keep the frame moving at constant speed})$$

and the work done by the applied force from $x = x_0$ to $x = 0$ is

$$W_{\text{app}} = \frac{4B^2 v x_0^3}{3R}.$$

The energy dissipated in the resistor over the same interval is

$$W_R = \frac{4B^2 v x_0^3}{3R},$$

so

$$W_{\text{app}} = W_R.$$

7. A metal crossbar of mass m slides without friction on two long parallel conducting rails, separated by a distance b , as shown in Fig. 3. A resistor R is connected across the rails at one end; compared with

R , the resistance of the bar and rails is negligible. There is a uniform magnetic field $\mathbf{B}$ perpendicular to the plane of the figure. At time $t = 0$ the crossbar is given a velocity v_0 toward the right.

What happens afterward?

- (a) Does the rod ever stop moving? If so, when?
- (b) How far does it go?
- (c) How about conservation of energy?

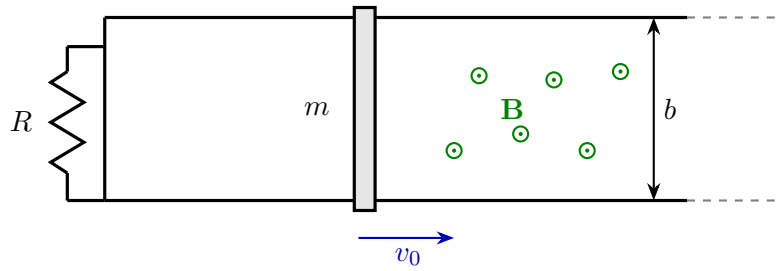
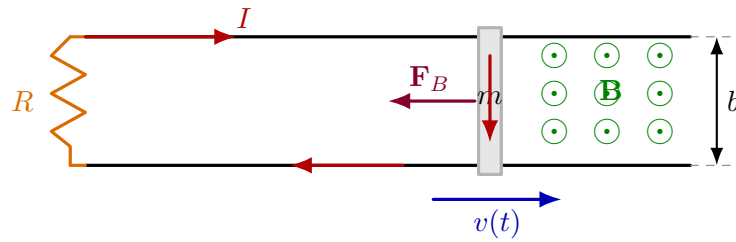


Figure 3: Sliding conducting bar on parallel rails in a uniform magnetic field.

Solution



1. Motional emf and induced current

As the bar moves to the right with speed v in a uniform magnetic field $\mathbf{B}$ perpendicular to the plane of the rails, the area of the loop changes. Therefore the magnetic flux changes, and a motional emf is induced.

The emf across the moving bar is

$$\mathcal{E} = B b v.$$

Since the resistance of the rails and the bar is negligible, the total circuit resistance is just R . Hence the induced current is

$$I(t) = \frac{\mathcal{E}}{R} = \frac{Bbv}{R}.$$

Direction of the current: The magnetic flux through the loop increases as the bar moves right, so by Lenz's law the induced current must oppose that increase. For the field direction shown in the figure, the current is clockwise, which makes the magnetic force on the bar point to the left, opposing the motion.

2. Magnetic force on the moving bar

The magnetic force on a straight conductor of length b carrying current I in a field B is

$$F_B = I b B.$$

Using $I = \frac{Bbv}{R}$,

$$F_B = \frac{Bbv}{R} bB = \frac{B^2 b^2}{R} v.$$

This force is opposite to the motion, so Newton's second law gives

$$m \frac{dv}{dt} = -\frac{B^2 b^2}{R} v.$$

Therefore,

$$\frac{dv}{dt} + \frac{B^2 b^2}{mR} v = 0.$$

Define the time constant

$$\tau = \frac{mR}{B^2 b^2}.$$

Then the equation becomes

$$\frac{dv}{dt} + \frac{1}{\tau} v = 0.$$

3. Velocity as a function of time

With the initial condition

$$v(0) = v_0,$$

the solution is

$$v(t) = v_0 e^{-t/\tau} = v_0 \exp\left(-\frac{B^2 b^2}{mR} t\right).$$

So the bar slows down exponentially.

4. Does the rod ever stop?

Strictly speaking, no. Since

$$v(t) = v_0 e^{-t/\tau},$$

the velocity approaches zero only asymptotically as $t \rightarrow \infty$. Thus,

$$\text{the bar never stops in finite time; } v(t) \rightarrow 0 \text{ only as } t \rightarrow \infty.$$

5. How far does it go?

Let $x(t)$ be the displacement of the bar from its initial position. Then

$$\frac{dx}{dt} = v(t) = v_0 e^{-t/\tau}.$$

Integrating from 0 to t ,

$$x(t) = \int_0^t v_0 e^{-t'/\tau} dt' \tag{29}$$

$$= v_0 \tau \left(1 - e^{-t/\tau}\right). \tag{30}$$

Hence,

$$x(t) = v_0 \tau \left(1 - e^{-t/\tau}\right).$$

As $t \rightarrow \infty$,

$$x(\infty) = v_0 \tau.$$

Therefore the total distance traveled is

$$x_{\max} = v_0 \tau = \frac{m R v_0}{B^2 b^2}.$$

6. Conservation of energy

The initial kinetic energy of the bar is

$$K_0 = \frac{1}{2} m v_0^2.$$

At time t , the kinetic energy is

$$K(t) = \frac{1}{2} m v^2(t).$$

The current in the resistor is

$$I(t) = \frac{B b v(t)}{R},$$

so the Joule-heating power is

$$P_R = I^2 R = \left(\frac{B b v}{R} \right)^2 R = \frac{B^2 b^2}{R} v^2.$$

Now differentiate the kinetic energy:

$$\frac{dK}{dt} = m v \frac{dv}{dt}. \quad (31)$$

Using the equation of motion

$$m \frac{dv}{dt} = - \frac{B^2 b^2}{R} v,$$

we get

$$\frac{dK}{dt} = - \frac{B^2 b^2}{R} v^2.$$

Therefore,

$$-\frac{dK}{dt} = P_R = I^2 R.$$

So the loss of kinetic energy of the bar is exactly equal to the electrical power dissipated in the resistor.

Integrating over all time,

$$\int_0^\infty I^2 R dt = \frac{1}{2} m v_0^2.$$

Thus all the initial kinetic energy is converted into heat in the resistor:

$$\frac{1}{2} m v_0^2 = \int_0^\infty I^2 R dt.$$

7. Current and force as functions of time

Since

$$v(t) = v_0 e^{-t/\tau},$$

we also have

$$I(t) = \frac{B b v_0}{R} e^{-t/\tau}$$

and

$$F_B(t) = \frac{B^2 b^2}{R} v_0 e^{-t/\tau}.$$

So the current and magnetic drag force also decay exponentially to zero.

Final Answers

The motion is governed by

$$m \frac{dv}{dt} = -\frac{B^2 b^2}{R} v.$$

Hence

$$v(t) = v_0 \exp\left(-\frac{B^2 b^2}{mR} t\right)$$

with time constant

$$\tau = \frac{mR}{B^2 b^2}.$$

(a) The rod never stops in a finite time:

$$v(t) \rightarrow 0 \text{ only as } t \rightarrow \infty.$$

(b) The displacement is

$$x(t) = v_0 \tau \left(1 - e^{-t/\tau}\right)$$

so the total distance traveled is

$$x_{\max} = v_0 \tau = \frac{mRv_0}{B^2 b^2}.$$

(c) Energy is conserved because the loss of kinetic energy equals the Joule heat:

$$-\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = I^2 R.$$

Thus,

$$\frac{1}{2} m v_0^2 = \int_0^\infty I^2 R dt.$$

8. A current I flows along a wire toward a point charge, causing the charge to increase with time. Consider a spherical surface S centered at the charge, with a tiny hole where the wire enters, as shown in Fig. 4. The circumference C of this hole is the boundary of the surface S .

Verify that the integral form of Maxwell's equation,

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a}, \quad (32)$$

is satisfied.

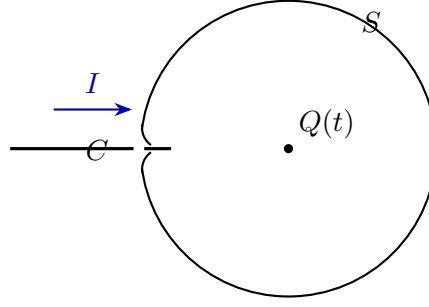


Figure 4: A spherical surface with a small hole through which the current-carrying wire enters.

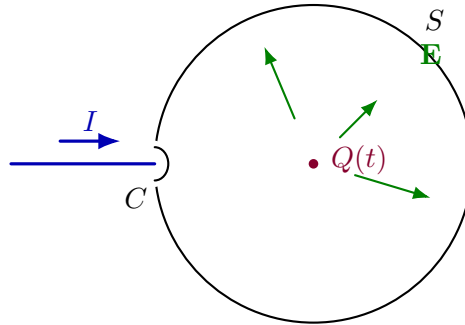
Solution

We want to verify, for the spherical surface S with a tiny hole,

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a}.$$

Here:

- a current I flows along the wire *toward* the point charge,
- the charge at the center is $Q(t)$,
- the spherical surface S has a small hole where the wire enters,
- C is the circular boundary of that hole.



1. Left-hand side: $\oint_C \mathbf{B} \cdot d\mathbf{s}$

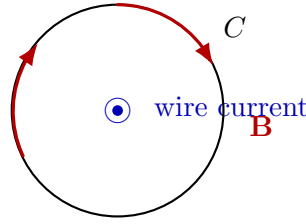
Around the straight wire, the magnetic field circles the wire according to the right-hand rule. If C is chosen with the orientation induced by the surface normal of S , then the magnetic field circulation around C is exactly the one corresponding to the current I entering through the hole.

Hence, by Ampère's law for a small loop linking the wire,

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \mu_0 I.$$

Therefore,

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \mu_0 I.$$



2. Right-hand side: conduction current term

Now consider

$$\iint_S \mu_0 \mathbf{J} \cdot d\mathbf{a}.$$

The current-carrying wire does *not* pierce the surface S ; it passes through the *hole*. So there is no conduction current through the actual surface S itself:

$$\iint_S \mathbf{J} \cdot d\mathbf{a} = 0.$$

Therefore the right-hand side reduces to the displacement-current term alone:

$$\iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a} = \mu_0 \epsilon_0 \iint_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a}.$$

3. Electric field flux through the spherical surface

The charge at the center is $Q(t)$, so the electric field is radially outward and, away from the tiny hole region, is essentially the Coulomb field:

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \frac{Q(t)}{r^2} \hat{\mathbf{r}}.$$

For a complete sphere centered on the charge, Gauss's law gives

$$\oiint \mathbf{E} \cdot d\mathbf{a} = \frac{Q(t)}{\epsilon_0}.$$

Since the hole is tiny, the flux missing through that tiny cap is negligible. Hence, to the required accuracy,

$$\iint_S \mathbf{E} \cdot d\mathbf{a} \approx \frac{Q(t)}{\epsilon_0}.$$

Differentiate with respect to time:

$$\iint_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a} = \frac{d}{dt} \left(\iint_S \mathbf{E} \cdot d\mathbf{a} \right) \approx \frac{d}{dt} \left(\frac{Q(t)}{\epsilon_0} \right) = \frac{1}{\epsilon_0} \frac{dQ}{dt}.$$

Therefore,

$$\mu_0 \epsilon_0 \iint_S \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a} = \mu_0 \frac{dQ}{dt}.$$

So the right-hand side becomes

$$\iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a} = \mu_0 \frac{dQ}{dt}.$$

4. Use charge conservation

Since the current I flows into the charge, the charge grows according to

$$\frac{dQ}{dt} = I.$$

Substituting this into the right-hand side gives

$$\iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a} = \mu_0 I.$$

Thus,

$$\iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a} = \mu_0 I.$$

5. Comparison both sides

From the left-hand side we found

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \mu_0 I,$$

and from the right-hand side we found

$$\iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a} = \mu_0 I.$$

Therefore,

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a}$$

as required.

Key idea

The wire current does not pass through the open spherical surface S , so the conduction-current term through S is zero. But the charge at the center is increasing, so the electric flux through S changes with time. That changing electric flux produces exactly the displacement-current term needed to make Maxwell–Ampère’s law work.

Final result

$$\oint_C \mathbf{B} \cdot d\mathbf{s} = \mu_0 I \quad \text{and} \quad \iint_S \left(\mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mu_0 \mathbf{J} \right) \cdot d\mathbf{a} = \mu_0 I.$$

Hence the integral form of Maxwell’s equation is satisfied.

9. Consider the magnetic flux density $\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{\phi}$ produced by the current I flowing along the x -axis.

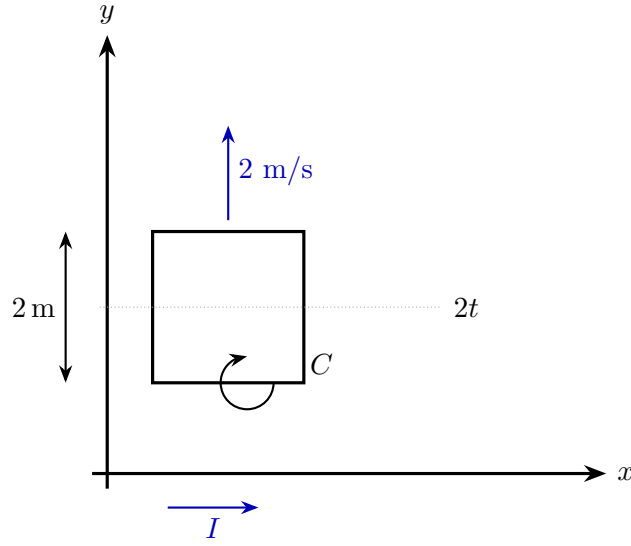


Figure 5: Square loop of area 4 m^2 moving upward in the magnetic field produced by a current-carrying wire along the x -axis.

What is the emf $\mathcal{E}$ of a square loop C of area 4 m^2 moving on the xy -plane with edges parallel to the x - and y -axes, if its center is located at $y = 2t$ m as a function of time? Compute the emf $\mathcal{E}$ first as

$$-\frac{d\Phi}{dt}$$

and then as

$$\oint (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

to verify that the same values are obtained.

Solution

We take the square loop to lie in the xy -plane, with edges parallel to the axes, and with center at

$$y_c = 2t.$$

Since the area is 4 m^2 , the side length is

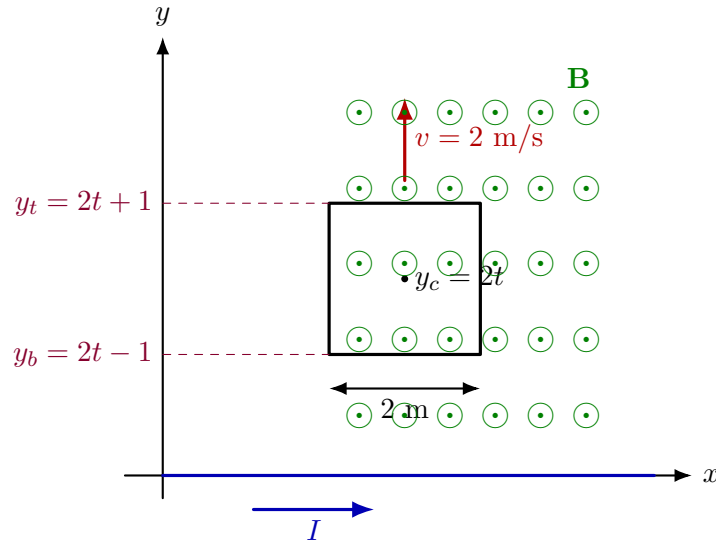
$$a = \sqrt{4} = 2\text{ m}.$$

Hence the loop extends from

$$x = -1 \text{ to } x = 1, \quad y = 2t - 1 \text{ to } y = 2t + 1.$$

We assume the loop stays entirely above the wire, so

$$2t - 1 > 0 \quad \implies \quad t > \frac{1}{2}.$$



1. Magnetic field in the plane of the loop

The current flows along the x -axis, so the magnetic field circles the x -axis. At points in the xy -plane with $y > 0$, the field points in the $+\hat{\mathbf{z}}$ direction (out of the page). Its magnitude is

$$B(y) = \frac{\mu_0 I}{2\pi y}.$$

Therefore, in the region of the loop,

$$\mathbf{B}(y) = \frac{\mu_0 I}{2\pi y} \hat{\mathbf{z}} \quad (y > 0).$$

2. emf from $-\frac{d\Phi}{dt}$

Take the positive area normal to be $+\hat{\mathbf{z}}$, so the positive loop direction is counterclockwise.

Since B depends only on y , the magnetic flux through the square is

$$\Phi(t) = \iiint \mathbf{B} \cdot d\mathbf{a} = \int_{x=-1}^1 \int_{y=2t-1}^{2t+1} \frac{\mu_0 I}{2\pi y} dy dx.$$

Because the integrand is independent of x ,

$$\Phi(t) = 2 \int_{2t-1}^{2t+1} \frac{\mu_0 I}{2\pi y} dy = \frac{\mu_0 I}{\pi} \int_{2t-1}^{2t+1} \frac{dy}{y}.$$

Thus,

$$\Phi(t) = \frac{\mu_0 I}{\pi} \ln\left(\frac{2t+1}{2t-1}\right).$$

Now differentiate:

$$\frac{d\Phi}{dt} = \frac{\mu_0 I}{\pi} \frac{d}{dt} \ln\left(\frac{2t+1}{2t-1}\right) \quad (33)$$

$$= \frac{\mu_0 I}{\pi} \left(\frac{2}{2t+1} - \frac{2}{2t-1} \right) \quad (34)$$

$$= \frac{\mu_0 I}{\pi} \left(\frac{2(2t-1) - 2(2t+1)}{(2t+1)(2t-1)} \right) \quad (35)$$

$$= -\frac{4\mu_0 I}{\pi(4t^2 - 1)}. \quad (36)$$

Hence the emf is

$$\mathcal{E} = -\frac{d\Phi}{dt} = \frac{4\mu_0 I}{\pi(4t^2 - 1)} \quad \left(t > \frac{1}{2}\right).$$

With our sign convention, this emf is *positive*, so the induced current is counterclockwise.

Result from flux

$$\mathcal{E}(t) = \frac{4\mu_0 I}{\pi(4t^2 - 1)}, \quad t > \frac{1}{2}.$$

3. emf from $\oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}$

Now verify the same answer using motional emf.

The loop moves upward with

$$\mathbf{v} = 2\hat{\mathbf{y}}.$$

Since

$$\mathbf{B} = \frac{\mu_0 I}{2\pi y} \hat{\mathbf{z}},$$

we have

$$\mathbf{v} \times \mathbf{B} = (2\hat{\mathbf{y}}) \times \left(\frac{\mu_0 I}{2\pi y} \hat{\mathbf{z}} \right) \quad (37)$$

$$= \frac{\mu_0 I}{\pi y} \hat{\mathbf{x}}. \quad (38)$$

So

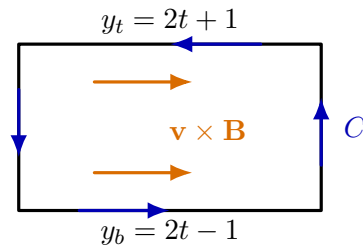
$$\mathbf{v} \times \mathbf{B} = \frac{\mu_0 I}{\pi y} \hat{\mathbf{x}}.$$

This points along $+\hat{\mathbf{x}}$. Therefore:

- on the two *vertical* sides of the loop, $d\mathbf{l} \parallel \hat{\mathbf{y}}$, so

$$(\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = 0;$$

- only the *top* and *bottom* horizontal sides contribute.



For the bottom side, $y = y_b = 2t - 1$, and $d\mathbf{l} = dx \hat{\mathbf{x}}$ with $x : -1 \rightarrow 1$. So

$$\mathcal{E}_{\text{bottom}} = \int_{-1}^1 \frac{\mu_0 I}{\pi(2t - 1)} dx \quad (39)$$

$$= \frac{2\mu_0 I}{\pi(2t - 1)}. \quad (40)$$

For the top side, $y = y_t = 2t + 1$, but the counterclockwise contour runs from $x = 1$ to $x = -1$, so

$$\mathcal{E}_{\text{top}} = \int_1^{-1} \frac{\mu_0 I}{\pi(2t + 1)} dx \quad (41)$$

$$= -\frac{2\mu_0 I}{\pi(2t + 1)}. \quad (42)$$

Therefore,

$$\mathcal{E} = \mathcal{E}_{\text{bottom}} + \mathcal{E}_{\text{top}} \quad (43)$$

$$= \frac{2\mu_0 I}{\pi(2t - 1)} - \frac{2\mu_0 I}{\pi(2t + 1)} \quad (44)$$

$$= \frac{2\mu_0 I}{\pi} \left(\frac{1}{2t - 1} - \frac{1}{2t + 1} \right) \quad (45)$$

$$= \frac{2\mu_0 I}{\pi} \left(\frac{(2t + 1) - (2t - 1)}{(2t - 1)(2t + 1)} \right) \quad (46)$$

$$= \boxed{\frac{4\mu_0 I}{\pi(4t^2 - 1)}}. \quad (47)$$

This is exactly the same result as before.

Verification by motional emf

$$\oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l} = \frac{4\mu_0 I}{\pi(4t^2 - 1)} = -\frac{d\Phi}{dt}.$$

4. Direction of the induced current

As the loop moves upward, the field magnitude $B(y) = \mu_0 I / (2\pi y)$ decreases. So the outward flux ($+\hat{\mathbf{z}}$) through the loop decreases with time.

By Lenz's law, the induced current must try to maintain an outward field. Therefore the induced current is

counterclockwise

when viewed from $+\hat{\mathbf{z}}$.

Final answer

Final Result

For the square loop of side 2 m whose center is at $y = 2t$, the emf is

$$\mathcal{E}(t) = \frac{4\mu_0 I}{\pi(4t^2 - 1)}, \quad t > \frac{1}{2}.$$

With the positive loop orientation taken as counterclockwise, this emf is positive, so the induced current is counterclockwise.

The same result is obtained both from

$$\mathcal{E} = -\frac{d\Phi}{dt}$$

and

$$\mathcal{E} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}.$$